

Yasah Sai Chandra Borusu

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EDUCATION

George Mason University, College of Engineering & Computing

Fairfax, VA

B.S. in Computer & Information Sciences

Aug 2022 – Dec 2026

Relevant coursework: Data Structures & Algorithms, Database Systems, Operating Systems, Machine Learning, Computer Networks, Software Engineering, Web Development, Probability & Statistics

EXPERIENCE

Software Engineer

Aug 2025 – Apr 2026

TheCollegeTech

Hybrid/Remote

- Implemented features across **Learning Management**, **Campus Management**, and **Placement Management** systems (attendance & gradebook, timetables, applications/shortlisting, notifications).
- Built reusable **React + TypeScript** component library (tables/forms/modals), form validation, and API clients with idiomatic hooks and state patterns.
- Designed **Python** REST endpoints with relational **SQL** data models (filtering, pagination, role-based access, parameterized queries, schema migrations); authored unit/integration tests and participated in PR reviews.

Cloud & AI Intern

May 2025 – Jul 2025

Astrani Technology Solutions (Ashburn, VA)

Hybrid/Remote

- Built **AI-driven business insights** tools on **AWS Bedrock** with **S3/DynamoDB** datasets; evaluated **OpenAI**, **DeepSeek**, and **Mistral** for CX analytics, summarization, and recommendations.
- Led **data engineering** migration from on-prem to **AWS RDS (PostgreSQL)** achieving nearly 70% infrastructure cost savings; designed ETL workflows and parameterized SQL for downstream reporting and analytics.
- Recommended domain-based **small-language-model (SLM)** strategy to reduce inference cost vs. large LLMs; supported image classification, text processing, and chatbot enhancements with experiment tracking.

Undergraduate Teaching Assistant (Python)

Aug 2023 – May 2026

College of Engineering & Computing, George Mason University

Fairfax, VA

- Guided **150+** students every semester during labs and office hours on Python fundamentals, debugging, and project best practices; delivered explanations and resources that improved comprehension and student project outcomes.

TECHNICAL SKILLS

Languages: Python, SQL (MySQL, PostgreSQL), Java, C/C++, TypeScript/JavaScript

Frameworks & Libraries: React, Next.js, Node.js, FastAPI, LangChain, PyTorch, OpenCV

Cloud & Databases: AWS (Bedrock, S3, DynamoDB, RDS), Microsoft Azure (Fundamentals), Supabase

Web: REST APIs, HTML/CSS/XML

Tools & OS: Git, Linux/Unix, MS Office (Word, Excel, PowerPoint)

Certifications: Microsoft Azure Fundamentals (AZ-900), Analytics Vidhya AI/ML Bootcamp

PROJECTS

BRIK-E: Smart-Building Hazard Detection — *Python, PyTorch, YOLOv8/SegFormer, OpenCV, FastAPI*

- Trained and deployed custom computer-vision models to detect spills, trash, and lost items on CCTV feeds; added a **two-stage verifier** (detector to classifier) to reduce false positives.
- Calibrated cameras to floor plans via **homography** for real-world (x, y) targets; built a real-time event pipeline over **MQTT** for robot dispatch and live tracking.

Automated Meeting Summarizer — *LangChain, Whisper, GPT-4/LLaMA-2, Google Calendar API*

- Live transcription with **Whisper**; automated extraction of key points, decisions, and action items into structured outputs.
- Integrated with **Trello/Asana** and **Google Calendar** APIs for downstream task creation, reminders, and deadline tracking.

MindEase: AI Task Planning Chatbot & Dashboard — *Next.js, TypeScript, Supabase (Postgres + Auth)*

- Built and deployed a chat-based productivity app that converts natural-language input into prioritized tasks (high/medium/low) and a daily action plan; supports multi-task capture per message, subtask breakdown, and a live dashboard with workload metrics.
- Architected **Next.js + TypeScript** frontend with **Supabase Postgres** backend and SSR auth; implemented **row-level security** policies, typed database contracts, and REST API routes for chat parsing, task CRUD, and dashboard summary.